Ilyas DAHAOUI *Data Scientist & AI Engineer*

dahaouiilyas@gmail.com -33745597285 Clermont FD, France portfolio GitHub
LinkedIn Hugging Face orcid.org/0009-0006-2945-5270

Comfortable in VUCA environments, I draw on a strong engineering background combined with expertise in AI, R&D, digital transformation, and consulting to deliver complex, high-impact projects. Available for a full-time position.

EXPERIENCE

03/2026 – 09/2026
Clermont-Ferrand, France. On-site

Michelin, *R&D Data Scientist Intern*
- Led grey-box model discovery for tire wear using advanced AI to identify governing dynamics. Built and validated models on large-scale time-series data, and developed a hybrid physics-informed deep learning framework for high-fidelity, real-time tire life cycle prediction, improving safety and operational performance.

04/2025 – 05/2026
Massy, France Remote

Nokia, *AI Research Scientist* (*Freelance*)
- Developed a novel deep learning model for telecommunication channel feedback prediction using Fourier Analysis Networks (FAN), attention mechanisms, and spatial feature modeling. Achieved a 90% performance improvement over state-of-the-art methods and earned acceptance for presentation at IEEE ICC 2026.

01/2025 – 02/2026
Canada, Remote

Université du Québec à Chicoutimi (UQAC), *AI Research Scientist* (*Freelance*)
- Designed an interpretable multimodal deep learning model (FoCLAP-Net) using focal modulation for wearable sensor time series posture recognition in rehabilitation. Achieved 98.66% classification accuracy with real time inference capability, improving clinical interpretability via modality aware attention mechanisms. Published in IEEE IoT: FoCLAP-Net.

09/2024 – 09/2025
Guyancourt, France. Hybrid

Ampere (Renault Group), *R&D Data Scientist & MLOps Intern*
- Developed and deployed a high-performance virtual sensor based on physics-informed thermal modeling and machine learning across the full lifecycle (R&D to vehicle integration), optimized for real-time inference and memory efficiency on industrial time-series data, achieving an MAE of ±1.XX°C ($R^2 > 0.9X$).:
- Ensured compliance with ISO 26262, ASIL B/C, and SOTIF while aligning technical design with business constraints, enabling a €15 per vehicle production cost reduction.

04/2024 – 07/2024
Lyon, France On-site

INSA Lyon CNRS/LIRIS, *AI Research Intern*
- Developed a hybrid Kolmogorov-Arnold Networks architecture for image deepfake detection, achieving 98.7% AUC and outperforming state-of-the-art methods by 6.6%.

07/2023 – 08/2023
Casablanca, Morocco On-site

Office National de l'Électricité et de l'Eau Potable (ONEE), *Machine Learning Intern*
- Led a predictive maintenance initiative for critical infrastructure, reducing unplanned downtime by over 40% through machine learning deployment.

EDUCATION

09/2022 – 03/2026
Casablanca, Morocco

École Centrale Casablanca CentraleSupélec Group, *Engineering Degree Data Science, Digitalization & Consulting*
- Generalist-oriented program focused on data science, AI, digital transformation, and consulting, developed in collaboration with OCP Solutions.

09/2024 – 09/2025
Paris, France

Université Paris Vincennes, *MSc in Artificial Intelligence Engineering* (*Work-Study Program*)
- Machine learning, deep learning, MLOps, generative AI, cloud computing, and cybersecurity.

01/2024 – 08/2024
Lyon, France

École Centrale Lyon, *Engineering Exchange Program*
- Machine learning, computer vision, signal processing, robotics, and optimization & collaborative algorithms.

09/2020 – 07/2022
Khouribga, Morocco

Classes Préparatoires aux Grandes Écoles (CPGE), *Mathematics & Physics Track*
- Advanced training in mathematics, physics, computer science, and engineering fundamentals.:

SKILLS SUMMARY

Soft Skills
Project management, Communication, Teamwork, Adaptability, Problem-solving, Languages: French (bilingual), English (fluent), Arabic (bilingual), Spanish (basic)..

Hard Skills
- ***Programming & Tools*:** Python, C/C++, R, SQL, MATLAB, EDA, feature engineering, data visualization (Power BI, Tableau, Looker); big data processing, SparQL.
- ***Artificial Intelligence*:** Supervised and unsupervised learning, deep learning, reinforcement learning, time series forecasting, NLP, computer vision, XAI, generative AI , RAG, fine-tuning (LoRA, PEFT), prompt engineering, AI agents, multi-agent systems.
- ***Frameworks & Libraries*:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, LangChain, LangGraph, LlamaIndex, AutoGen; FastAPI, Flask, Streamlit.
- ***MLOps / LLMOps*:** MLflow, Airflow, TFX; Docker, Kubernetes, CI/CD pipelines, Git; monitoring tools (Prometheus, Grafana).
- ***Cloud & Big Data*:** AWS, Azure, GCP; Databricks, Spark, Snowflake, BigQuery; ETL pipelines and data engineering workflows.
- ***Databases*:** PostgreSQL, MongoDB, vector databases (FAISS, Chroma, pgvector).
- ***Mathematics & Statistics*:** Probability, statistics, causal inference, linear algebra, optimization, hypothesis testing, time-series analysis, decision theory, graph theory, linear programming.
- ***Certifications*:** Master Retrieval-Augmented Generation (RAG) Systems – Packt (2025), MCP– Hugging Face (2025), Agents AI – Hugging Face (2025), Time Series Analysis with Python & Statistical Tests in Sales and Marketing– 365 Data Science (2024),....

PROJECTS

AI ATS Matching Pipeline Forvis Mazars,
Big data pipeline for CV ingestion (OCR), normalization, indexing, and LLM-based candidate-job matching with bias mitigation and prompt-injection robustness.
Tech: LLMs, NLP, OCR, Data Engineering, Python

REA App Multi-Agent Research Assistant, *Multi-agent LLM system for automated scientific report generation combining arXiv and web sources (Tavily). Implements tool-augmented LLM reasoning, summarization, and trend analysis with Streamlit interface.*
Tech: LLMs, Multi-agent systems, OpenRouter, arXiv API, Python, Streamlit

Advanced RAG System Hybrid Retrieval Pipeline, *End-to-end RAG system with hybrid retrieval (ChromaDB BM25), query rewriting (HyDE, Multi-Query, Self-Ask), and intelligent routing for adaptive search strategies.*
Tech: RAG, LLMS, Embeddings, ChromaDB, BM25, OpenRouter, Python

CobAI - AI Powered COBOL System Evolution, *Developed CobAI to automate COBOL code generation, legacy code conversion, and documentation, while optimizing system performance and energy efficiency using AI on synthetic data.*
Tech: Python, AI, Natural Language Processing, COBOL, Mainframe Integration, Synthetic Data

Vulnera Scan LLM Security Tool, *LLM-based vulnerability detection system fine-tuned with GRPO reinforcement learning, achieving 85% accuracy on the Google CodeXGLUE defect detection dataset.*
Tech: GRPO, RL, LLMs, LoRA, QLoRA

AI Real Estate Chatbot Pzartech Ltd, *Conversational search engine replacing traditional filters with an LLM-powered interface over scraped real estate databases, improving user experience and search efficiency.*
Tech: LLMs, NLP, Web Scraping, Python

Bitcoin Fraud Detection Graph Neural Networks, *Fraud detection on Elliptic dataset using Graph Neural Networks with graph feature engineering (centrality, communities) and semi-supervised learning, improving recall via unlabeled node handling.*
Tech: GNNs, PyTorch, Graph ML

AI Negotiation Assistant (MIT x UN Hackathon),
Fine-tuned Qwen-3-14B with RLIHF to generate structured negotiation outputs, improving workflow efficiency for humanitarian conflict resolution (+35%).
Tech: Transformers, RLHF, TRL, fine-tuning

AeroNavBot - AI Assistant for Aviation Navigation and Flight Planning, *AI aviation assistant providing real-time navigation, weather, and flight calculations to improve planning accuracy, decision-making, and safety, with future integration of global flight management tools.*
Tech: Python, LLM, Agents, Weather APIs, Flight Data Integration

VOLUNTEER EXPERIENCE

03/2026 – Present	**MoroccoHPC,** *Partnership & Sponsorship/Technical Content Member* Member of the national high-performance computing community focused on advancing scientific computing and AI infrastructure in Morocco.
09/2025 – Present	**AI4Impact Spring School (École Polytechnique, MBZUAI, UM6P),** *Organizing Committee Member* Led organization of the AgorAI Hackathon, including jury participation and coordination of student AI-for-impact projects.: Contributed to strategic outreach and partnerships for a high-level scientific event featuring Yann LeCun and Eric Xing on World Models.:
07/2025 – 01/2026	**Hack-Nation Global AI Hackathon,** *AI Ambassador* Expanded global participation and community engagement for one of the largest AI hackathons, supporting outreach, events, and mentoring initiatives.
09/2024 – Present	**Quantum Computing and AI Consultant,** *Centrale Quanta* Providing consultancy on the integration of AI techniques with quantum computing to enhance research and application potential within both fields.
02/2024 – Present	**Quantum Physics & Quantum Computing Advocate,** *Mail & Maroc* Advocating for the integration of quantum computing into educational frameworks and research initiatives in Morocco, raising awareness of the benefits of quantum technologies.
10/2023 – 10/2024	**Research Staff Member,** *Moroccan Mathematical Community* Contributing to the development and research in the field of quantum physics, focusing on mathematical modeling and research advancement within the community.
09/2022 – 2025	**Volleyball Player,** *Bureau des sports — Centrale Lyon-Centrale Casablanca* Member of the volleyball team, representing École Centrale Casablanca in multiple university sports events and promoting team spirit and fitness within the campus.

PUBLICATIONS

FoCLAP-Net: Toward an Interpretable Model for Body Posture Recognition via Multimodal Temporal Data From Wearable Sensors , *IEEE Internet of Things Journal*

Deep Learning for Massive MIMO CSI Prediction Using Empirical Data, *IEEE International Conference on Communications* 2026